



AWS Certified Cloud Practitioner

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Cloud Concepts

Cloud Computing

It refers to use of a network of remote servers to store, manage and process data instead of using local or personal computers. Earlier there were multiple ways to host an application on web:

- Self host → refers to using a personal/on premise/local computer system to host an application where you buy the components, hire professionals for managing the infra, pay for cost of real estate and handle all the risks associated with it. A single system was often use for a single business application or purpose which leads to under utilization of resources
- VPS(virtual private servers) → virtualization enables a single machine to be used for multiple purposes to host application and enables better utilization of resources
- Shared hosting → shared hosting lets hundreds of businesses to host their applications on a single machine often using virtualization or dedicated storage with permissions e.g. godaddy, but a single process of a business may lead to excess usage of server affecting other business owners
- Cloud providers → multiple machines which acts as a single system that is scalable, flexible and configurable. You only need to configure your

infrastructure and don't need to worry about physical components or resources limits

Cloud Providers

A cloud provider is a company that may fulfil most or all of the below requirements:

- offers multiple or a set of cloud services
- those cloud services can be chained together to create a cloud architecture
- those services can be accessible via a single API endpoint
- metered billing or usage based billing is enabled on those services
- it provides infrastructure as a service (IaaS) with network and storage components
- it provides support for infrastructure as code or IaC
- provides monitoring support for all of the cloud services

AWS is currently the leading cloud provider started in 2004 with first service **SQS(simple queue service)** being released and then **S3(simple storage service)** at second and then **EC2** at third

Common Cloud Services

- Compute
- Networking
- Storage
- Database

Benefits of Cloud

- Agility → speed, agile
- Cost effective → Pay as you go billing/on demand
- Economy of scale

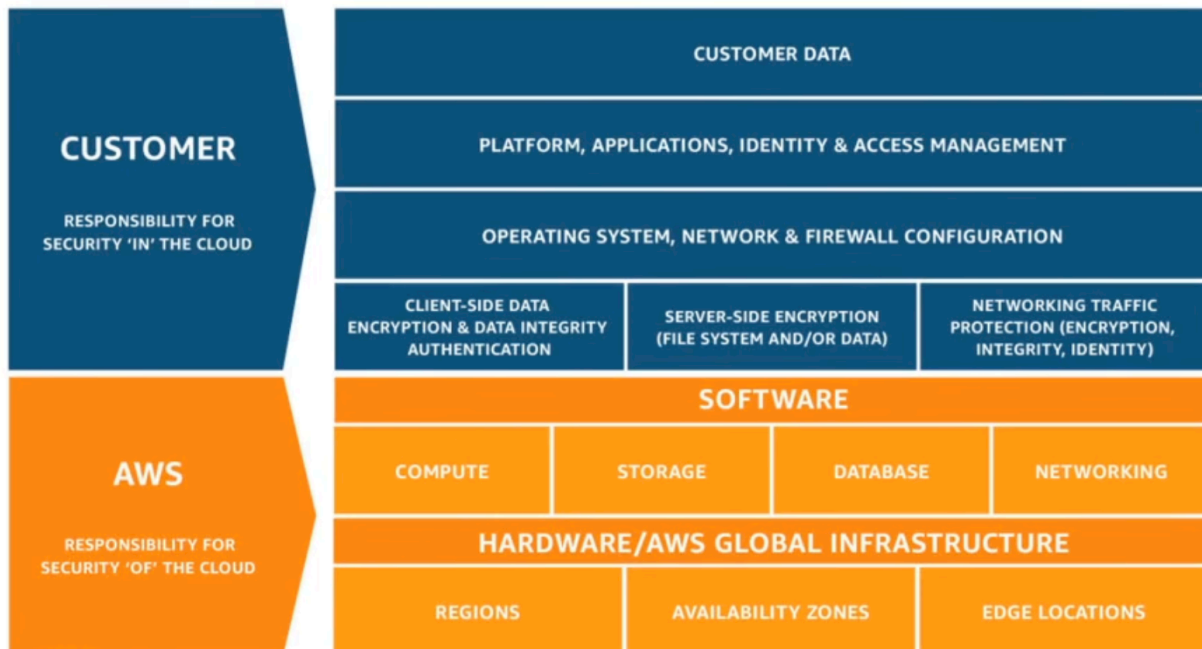
- Global reach in minutes → launch workloads anywhere in the world
- Security → AWS infrastructure is secure by default and you can make the access granular
- Reliability → data backup, disaster recovery, fault tolerance
- Scalability → increase or decrease loads as per your needs easily
- High Availability
- Elasticity → automate scaling during spikes
- Current → underlying software and hardware is patched, upgraded and licensed by cloud provider without you needing to worry about

Types of cloud computing

- On premise → you manage everything from hardware, storage, networking, servers, OS, runtime, virtualisation, applications and data along with maintenance
- Infrastructure as a service (IaaS) → The hardware, storage, networking, compute and maintenance of it is managed by CSP whereas you manage OS, runtime, application and data. Examples are AWS, azure, GCP
- Platform as a service (PaaS) → You only manage the application code and user/application data whereas all the hardware, OS and runtime configuration is managed. e.g. Vercel, Railway
- Software as a Service (SaaS) → you only care about getting your work done and don't care about anything like application code data, runtime, servers etc. e.g. Adobe creative cloud, social sites

Shared responsibility model

It is a cloud security framework that defines security obligations of customer vs cloud service provider. Customers are responsible for security in the cloud whereas AWS is responsible for security of the cloud. Think of it like if you can configure something yourself then you, the customer is responsible for it whereas if you are not able to configure anything then CSP is responsible for that resource



Computing services

EC2

Elastic compute service allows you to launch virtual machines. VM is a software that lets you emulate a physical computer using virtualisation. It allows you to create, copy, resize and migrate your server easily. EC2 is highly configurable where you can select amount of CPU, RAM, storage options, network bandwidth and guest OS. Multiple VMs can run on the same physical server with host OS. Amazon machine image (AMI) lets you create a template or configuration of your server that you can reuse multiple times. A running VM is known as instance.

Lightsail is a managed virtual server service or a friendly version of EC2

Containers

They are lightweight VMs can use the guest OS on a virtual machine or host OS if they are running on bare metal EC2. Containers allow better utilisation of underlying hardware or VM by running multiple workloads inside separate containers on same VM. **Elastic container service (ECS)** provides provisioning of containers by first creating a cluster.

Even if a container is not in the running state the ECS will start charge you for the VM that is provisioned to run the container if it started, **ECS Fargate** is used to as serverless ECS that will only charge for the running containers.

Elastic container registry (ECR) acts as a repository to store your container images

Elastic kubernetes service (EKS) is a managed kubernetes service by AWS

Serverless

They are functions of piece of code that runs for the amount of time you specified and consume memory that you configure. You don't need to configure a VM or setup a runtime to run an application instead you handover your code and select a runtime and AWS configures a VM and runtime for you can only charge for the amount of time the code runs rounded to 100 milliseconds. **AWS Lambda** functions provide you the functionality to create serverless functions

Cost and capacity management

Cost management is how do we save money and capacity management is about how do we meet the demands of traffic by upgrading or adding servers

AWS spot, reserved and savings plans is about paying upfront in full or partially or make a contract about upfront resources to meet the demands by high availability

AWS batch plans, schedule and execute batch workloads, AWS compute optimizer saves costs by using ML algorithms from your past usage, EC2 autoscaling groups (ASG) reduces costs by scaling up and down servers based on traffic, Elastic load balancer routes traffic to healthy EC2 instances even in different availability zones, Elastic bean stalk (EB) configures your runtime and deploy your code without you explicitly creating and managing infra which is similar to railway or vercel

Storage Services

Types of storage

Elastic block storage (EBS) → attaches a virtual hard drive to EC2 instance or server and operating system can directly access the drive and write to it using protocols like FC

Elastic file storage (EFS) → uses protocols like NFS and stores file and its metadata with multiple reads writes and file locks. Multiple devices can connect to the same EFS instance so use it when you need multiple VMs to have same storage layer or need to share files between multiple servers

Simple storage service (S3) → saves file with meta data and unique id. It is a managed service for storing and accessing file where you just upload and read using unique id, not intended for high IOPs, doesn't support file locks and using HTTPS protocol

Object storage and S3 storage classes

AWS S3 stores files as objects instead of just files in a hierarchy similar to a filesystem(elastic file system) or as blocks of data on a hard drive(elastic block storage) so each object has key(unique name or id), value(sequence of bytes holding data), version id(if versioning is enabled) and meta data. S3 has below storage classes which trades retrieval time, accessability, durability and cost

- S3 standard(default) → fast, 99.9% availability, 11 9's durability and three AZs replication
- S3 intelligent tiering → uses ML to identify the right cost effective storage class based on file and object usage
- S3 standard IA(infrequent use) → fast and cheaper if file is accessed once a month otherwise it will cost more than S3 standard for retrieval
- S3 one zone IA → similar to standard IA but cheaper due to no replication so data is stored in only one AZ so there is high chance of data loss or low durability
- S3 glacier → data retrieval may take minutes to hours but storage is very cheap
- S3 glacier deep archive → cheapest tier option but data retrieval takes 12 hours

AWS Backup is a managed backup services for multiple AWS services like EC2, dynamoDB etc

Cloud Endure is disaster recovery service that continuously replicates your data to staging area